

CS101: Introduction to Artificial Intelligence

Course Syllabus - Fall Semester

Welcome to CS101! This course provides a broad introduction to the fundamental concepts of Artificial Intelligence (AI). We will explore the history of AI, search algorithms, knowledge representation, and the ethical implications of automation in modern society.

Course Objectives:

- Understand the core differences between human intelligence and machine intelligence.
- Master search state-space algorithms, including Breadth-First Search (BFS) and Depth-First Search (DFS).
- Develop a basic understanding of neural networks and deep learning models.
- Analyze the societal, economic, and ethical impacts of deployment of AI systems.

Grading Policy and Evaluation

Your final grade in this course will be calculated based on your performance in practical programming assignments, quizzes, a mid-term exam, and a final group project.

Grade Distribution:

- **Programming Assignments (40%):** There will be four hands-on Python coding assignments throughout the semester. Each assignment is worth 10% of your final grade.
- **Quizzes (15%):** Weekly short online quizzes to test your understanding of the reading materials.
- **Mid-term Examination (20%):** A written exam covering conceptual foundations and algorithms.
- **Final AI Project (25%):** Working in groups of 3, students will build and deploy a basic machine learning application using Python.

Late Work Policy:

Late assignments will face a penalty of 10% deduction per day. No submissions will be accepted more than 5 days past the original deadline.

Weekly Schedule and Readings

- **Week 1-2: Introduction to Classical Search**

We will study informed and uninformed search strategies. Readings: Chapter 1 and 2 of the course textbook.

- **Week 3-4: Knowledge Representation**

Understanding propositional logic, first-order logic, and semantic networks. Readings: Chapter 3.

- **Week 5-7: Machine Learning Foundations**

An introduction to supervised learning, linear regression, decision trees, and basic clustering algorithms. Readings: Lecture notes and assigned research papers.

- **Week 8: Mid-term Exam**

Held in-class during the regular lecture hour. Covers all topics from Week 1 to Week 7.